



Mobile Application System Design

(Chap 3 – Part 2)

Course Lecturer: Nurul Amera Muhamad Nazmi

Key Concepts:

1. Widgets in UI
2. Styling (color, size, alignment)
3. Images & Icons
4. Text & TextField
5. ListView

Widgets in UI

Visible widgets – Text, Image, Icon, Button, TextField

Layout/structural widgets – Scaffold, Container, Center, Row, Column, ListView, Stack

Categories of common UI widgets

A. Structure and screen

- MaterialApp: the root of your app
- Scaffold: page layout with appBar, body, floating action button, drawer

B. Layout and positioning

- Container: a box that can style and hold a child
- Center: centers one child
- Row and Column: place children horizontally or vertically
- ListView: scrollable list for many items
- Stack: overlap children (e.g., badge over an image)

C. Content widgets

- Text
- Image
- Icon
- TextField: user

D. Interaction Widget

- ElevatedButton
- IconButton
- GestureDetector

Styling: Color, Size, and Alignment

1. Color – Color is what gives your app personality. It can make the screen cheerful, elegant, or exciting depending on what colors you choose

For example: `Colors.red`, `Colors.blue`

2. Size – Controls how big or small each element looks

3. Alignment – Controls where things appear on the screen

For example:

`Center` → puts content in the middle.

`Alignment.topCenter` → moves content to the top center.

`Alignment.bottomRight` → moves content to bottom-right corner.

Images & Icons in Flutter UI

Images

1. Why Images and Icons Matter in UI

- More engaging
- Easier to navigate
- More memorable to users

2. Images from your app (assets)

- Using pubspec.yaml (file paths under 'assets')

3. Images from the internet

- Flutter can load images from a URL link

4. How to Style Images

- Width and height : to resize the image
- Fit : to decide how the image fills its space (cover, contain, fill)
- Alignment : to position it in the container
- Border radius : to make rounded corners (like a food card)

Icons

1. Small symbols (represent actions / categories)

- 🏠 for home
- ❤️ for favorite

2. Material Icons (Flutter built-in icon library)

- Icons.home
- Icons.search
- Icons.favorite
- Icons.shopping_cart

3. Custom Icons (from SVG files)

- Using a package called flutter_svg

Text & TextField in Flutter UI

Text

1. Text Widget : Displaying Information

- show static text: words that you want the user to read but not edit

2. Styled using

- Font size
- Font weight (bold/light)
- Color
- Alignment

TextField

1. TextField Widget : Taking Input from Users

- Search bars
- Login Forms
- Feedback / Comment boxes

2. Styled using

- decoration : adds a border, hint text, or icons
- fillColor : background color
- contentPadding : space inside the box

3. Custom Icons (from SVG files)

- Using a package called flutter_svg

Listview in Flutter UI

Listview

1. What is ListView?

- A layout widget that can hold many children widgets which automatically adds scrolling when the content is too long to fit on one screen

2. Why Use ListView?

- Lets users scroll up and down easily.
- Makes the layout responsive on small screens.
- Works perfectly for showing repeated patterns (i.e: like a list of dishes, prices, or icons.)

3. How It Works

- A ListView is like a long column of items but with scroll enabled automatically

4. Common Mistakes

- Forgetting to limit ListView height inside a Column
- Forgetting padding
- Using too many large images

CLASS ACTIVITY

(Hands-On Practice)

Content

1. User Interface Design

1. Stateless and Stateful Widget

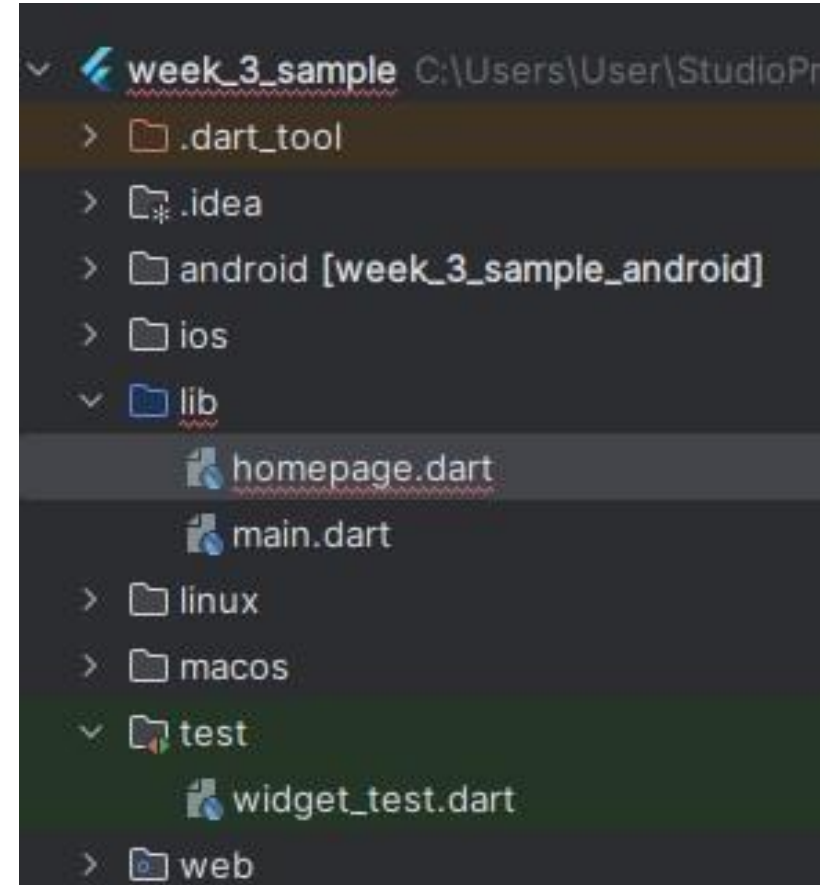
2. UI Elements and Stylings

- Text in Mobile Application
- Dimensions and Sizes
- Colors
- Images and Icons

- The basic building block of a Flutter App interface is a **widget**.
- Two main types of widgets
 1. Stateless Widget
 2. Stateful Widget
- A stateful widget can be nested within a stateless widget

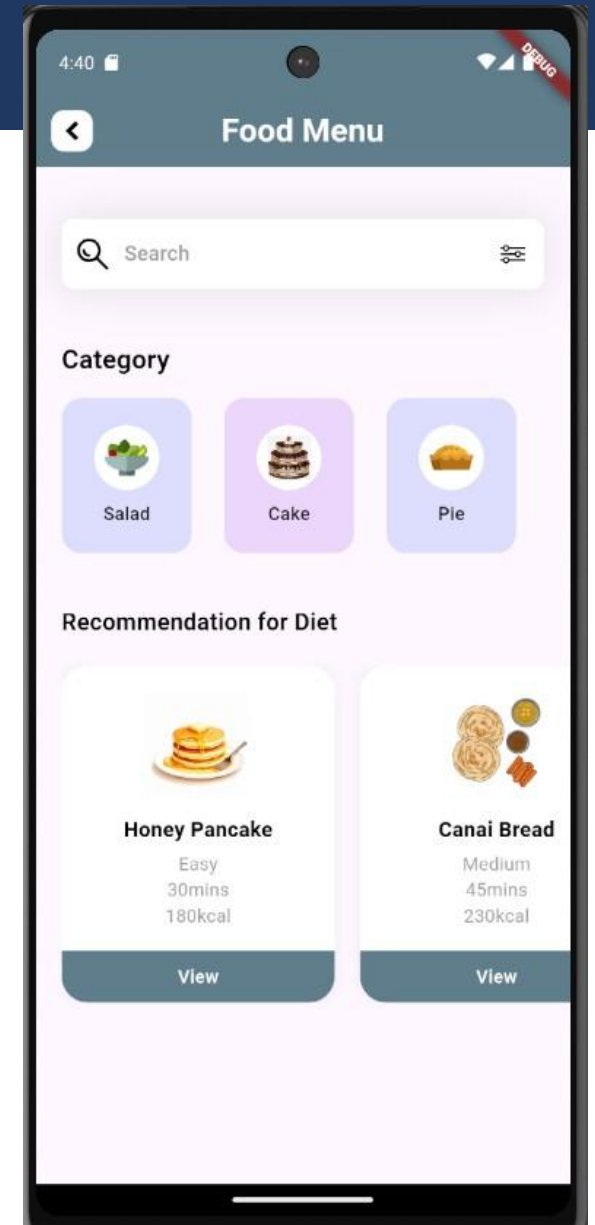
Hands-On – Restaurant App

- Create a new Flutter project.
- Create a new Dart file named “homepage” in the Flutter lib folder.



Hands-On – Restaurant App

- Today, we will build a home screen for a restaurant app.
- You will learn how to create the following widgets:
 - ✓ Text and TextFields
 - ✓ Image and Icon
 - ✓ ListView
- You will also learn in more detailed the way to set the alignment, dimension and color for your widgets.



Hands-On – Restaurant App

STEP 1: Create a New Flutter Project

- lib/
 - └─ mainrestaurant.dart
 - └─ homepage.dart
 - └─ models/
 - └─ category.dart
 - └─ diet.dart
- mainrestaurant.dart will be our starting point (where main() is).
- homepage.dart will be the main screen design.
- models/ folder will store the data models (Category and Diet info).
- Import all assets to your project (images / icons)
- Update assets for pubspec.yaml

```
import 'package:flutter/material.dart';  
import 'package:practice/homepage.dart';
```

```
Run | Debug | Profile  
void main() {  
  runApp(  
    const MaterialApp(  
      home: HomePage(),  
    ), // MaterialApp  
  );  
}
```

Hands-On – Restaurant App

STEP 2 : In homepage.dart, just type this for now:

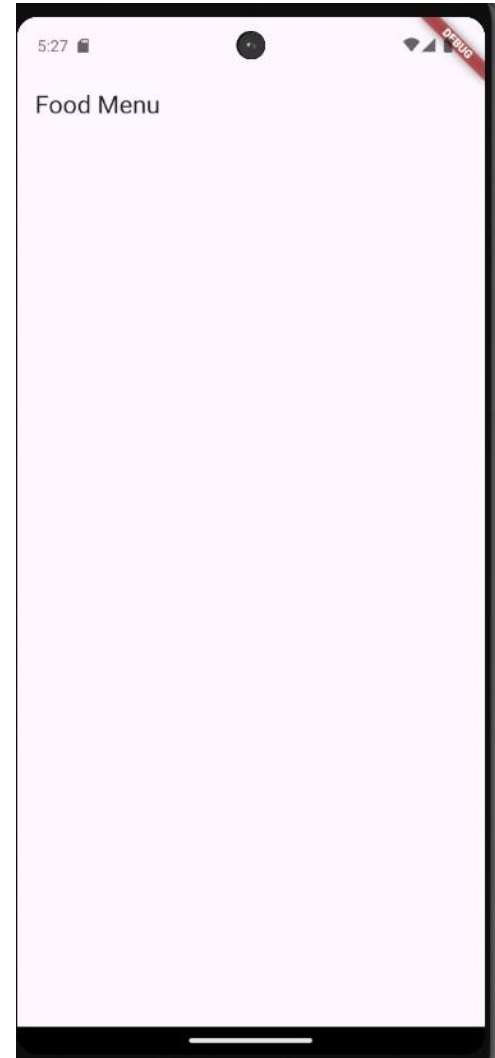
👉 Open homepage.dart.

- Creates a screen (Scaffold) with an AppBar showing "Food Menu" title.
- No body yet.

```
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Food Menu"),
      ), // AppBar
    ); // Scaffold
  }
}
```



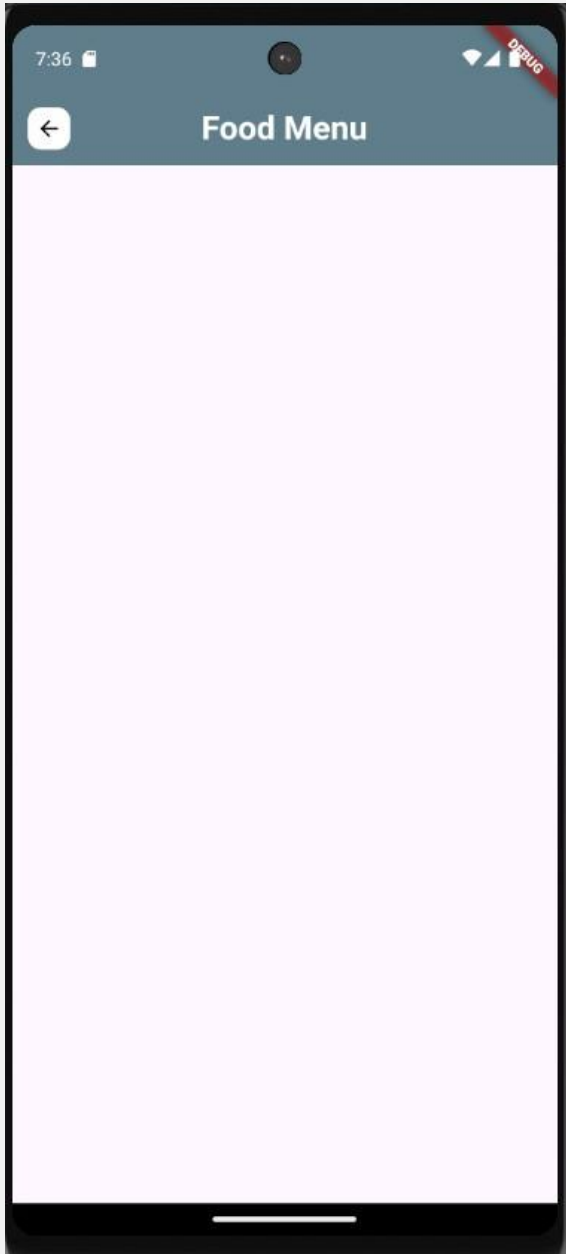
Hands-On – Restaurant App

STEP 3 : Build the AppBar and Search Bar (Basic UI)

👉 In homepage.dart.

- First, instead of a plain AppBar, we create a custom function for AppBar to make it clean and reusable.
- Then, in your Scaffold, change:
`appBar: _appBarFunction(),`

```
AppBar _appBarFunction() {  
  return AppBar(  
    title: const Text(  
      "Food Menu",  
      style: TextStyle(  
        color: Colors.white,  
        fontSize: 25,  
        fontWeight: FontWeight.bold,  
      ), // TextStyle  
    ), // Text  
    backgroundColor: Colors.blueGrey,  
    centerTitle: true,  
    leading: Container(  
      margin: const EdgeInsets.all(12),  
      alignment: Alignment.center,  
      decoration: BoxDecoration(  
        color: Colors.white,  
        borderRadius: BorderRadius.circular(10),  
      ), // BoxDecoration  
      child: const Icon(  
        Icons.arrow_back,  
        color: Colors.black,  
        size: 20,  
      ), // Icon  
    ), // Container  
  ); // AppBar  
}
```



STEP 4 : Add a Search Bar Below the AppBar

👉 In homepage.dart.

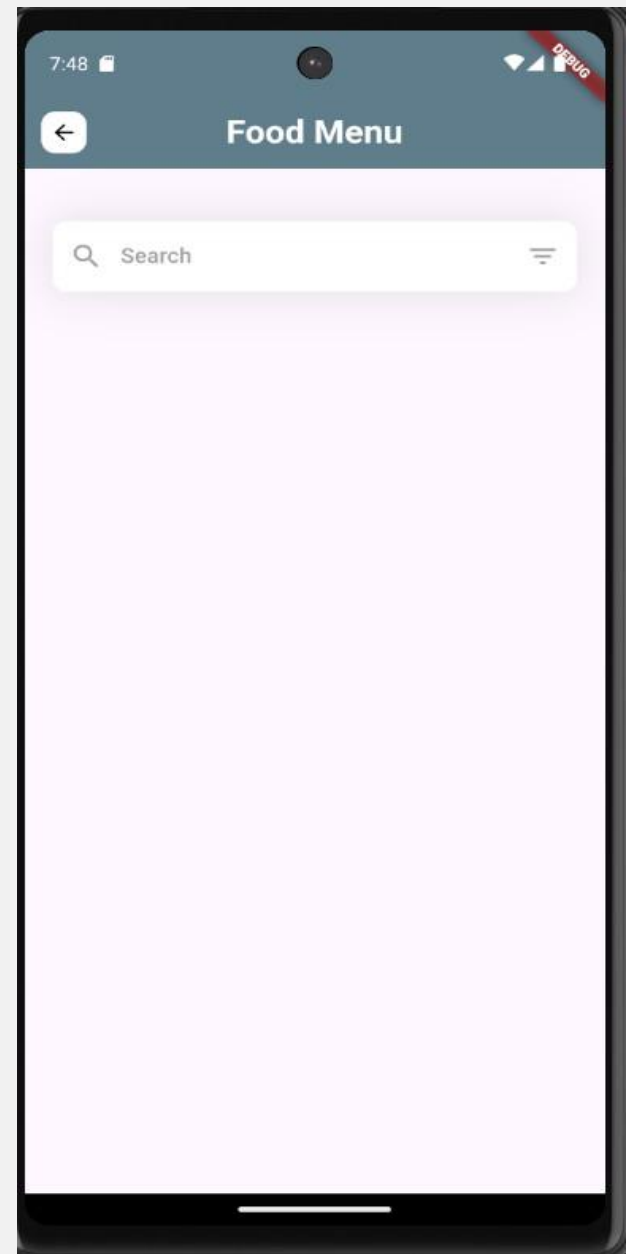
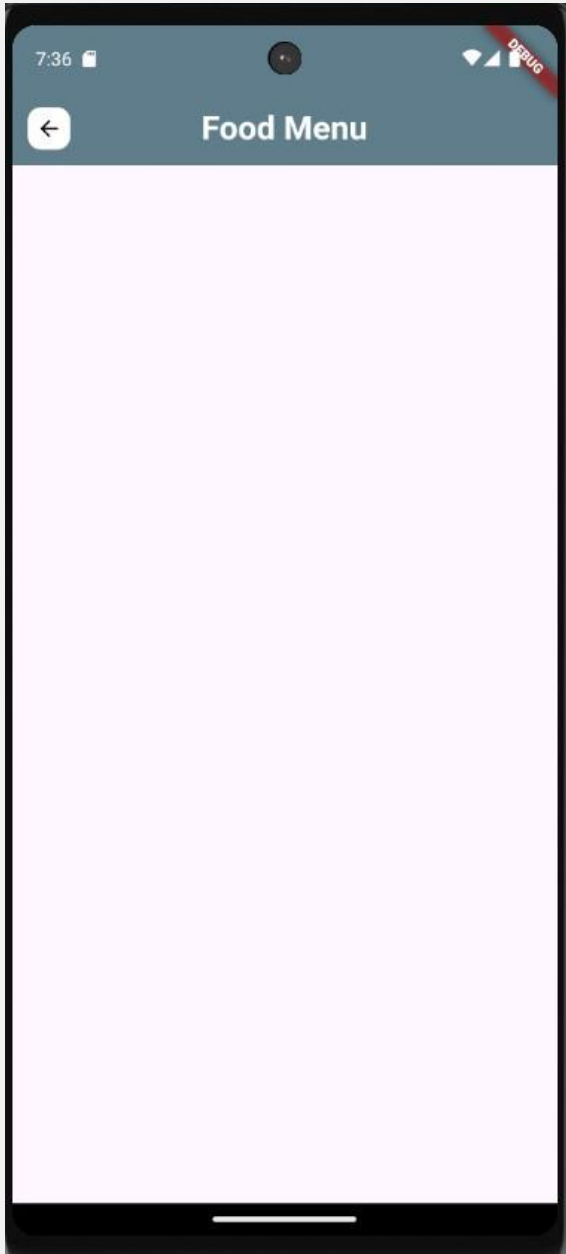
- Next, add a search bar to let users "search" menu items (even though not functional yet, it's for design).
- Below your `_appBarFunction()`, create another function
- Then inside `build()`, update your body

```
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: _appBarFunction(),
      body: SingleChildScrollView(
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            _searchField(), // add here
          ],
        ), // Column
      ), // SingleChildScrollView
    ); // Scaffold
  }
}
```

```
Container _searchField() {
  return Container(
    margin: const EdgeInsets.only(top: 40, left: 20, right: 20),
    decoration: BoxDecoration(
      boxShadow: [
        BoxShadow(
          color: Colors.black12.withOpacity(0.10),
          blurRadius: 30,
        ), // BoxShadow
      ],
    ), // BoxDecoration
    child: TextField(
      decoration: InputDecoration(
        filled: true,
        fillColor: Colors.white,
        contentPadding: const EdgeInsets.all(15),
        hintText: "Search",
        hintStyle: const TextStyle(
          color: Colors.grey,
          fontSize: 16,
        ), // TextStyle
        border: OutlineInputBorder(
          borderRadius: BorderRadius.circular(10),
          borderSide: BorderSide.none,
        ), // OutlineInputBorder
        prefixIcon: const Padding(
          padding: EdgeInsets.all(12.0),
          child: Icon(Icons.search, color: Colors.grey),
        ), // Padding
        suffixIcon: const Padding(
          padding: EdgeInsets.all(12),
          child: Icon(Icons.filter_list, color: Colors.grey),
        ), // Padding
      ), // InputDecoration
    ), // TextField
  ); // Container
}
```



STEP 4 : Create the CategoryModel Class

- 👉 In lib/models/, open category.dart file.
- Copy from category.txt and paste it there

```
import 'package:flutter/material.dart';

class CategoryModel {
  final String name;
  final String iconPath;
  final Color boxColor;

  CategoryModel({
    required this.name,
    required this.iconPath,
    required this.boxColor,
  });

  static List<CategoryModel> getCategories() {
    List<CategoryModel> categories = [];

    categories.add(
      CategoryModel(
        name: "Salad",
        iconPath: "assets/icons/salad.svg",
        boxColor: const Color(0xff92A3FD),
      ), // CategoryModel
    );

    categories.add(
      CategoryModel(
        name: "Cake",
        iconPath: "assets/icons/cake.svg",
        boxColor: const Color(0xffC58BF2),
      ), // CategoryModel
    );

    categories.add(
      CategoryModel(
        name: "Pie",
        iconPath: "assets/icons/pie.svg",
        boxColor: const Color(0xff92A3FD),
      ), // CategoryModel
    );

    return categories;
  }
}
```

STEP 5 : Create the DietModel Class

- 👉 In lib/models/, open diet.dart file
- Copy from diet.txt and paste it there

```
import 'package:flutter/material.dart';

class DietModel {
  final String name;
  final String iconPath;
  final String level;
  final String duration;
  final String calorie;
  final Color boxColor;

  DietModel({
    required this.name,
    required this.iconPath,
    required this.level,
    required this.duration,
    required this.calorie,
    required this.boxColor,
  });

  static List<DietModel> getDiets() {
    List<DietModel> diets = [];

    diets.add(
      DietModel(
        name: "Honey Pancake",
        iconPath: "assets/images/honey-pancake.jpg",
        level: "Easy",
        duration: "30mins",
        calorie: "180kcal",
        boxColor: const Color(0xff92A3FD),
      ), // DietModel
    );

    diets.add(
      DietModel(
        name: "Canai Bread",
        iconPath: "assets/images/canai-bread.jpg",
        level: "Medium",
        duration: "45mins",
        calorie: "230kcal",
        boxColor: const Color(0xffC58BF2),
      ), // DietModel
    );

    return diets;
  }
}
```

Hands-On – Restaurant App

Search flutter svg (using browser)

flutter_svg 2.2.1 

Published 38 days ago •  flutter.dev

SDK | FLUTTER | PLATFORM | ANDROID | IOS | LINUX | MACOS | WEB | WINDOWS

 5.7K

Readme | Changelog | Example | Installing | Versions | Scores

Use this package as a library

Depend on it

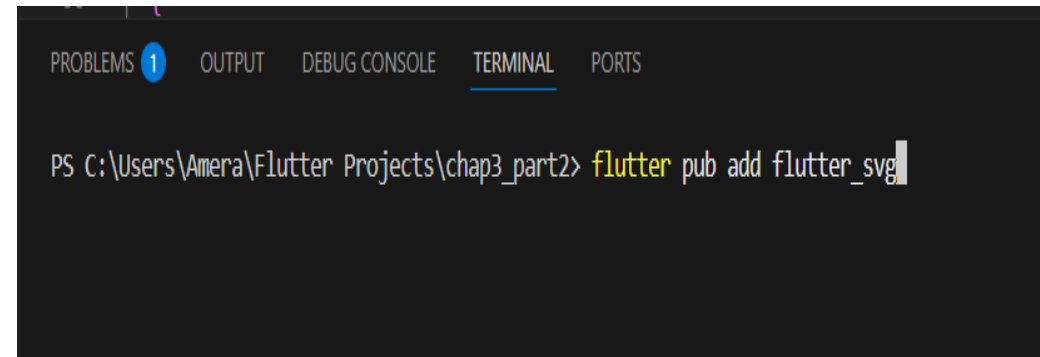
Run this command:

With Flutter:

```
$ flutter pub add flutter_svg
```

This will add a line like this to your package's pubspec.yaml (and run an implicit `flutter pub get`):

```
dependencies:  
  flutter_svg: ^2.2.1
```



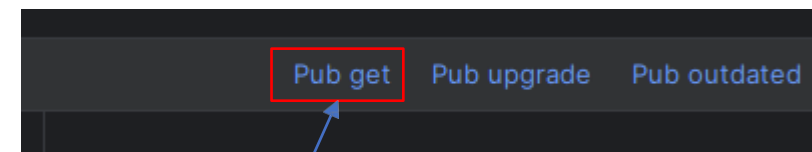
```
PROBLEMS 1 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS  
  
PS C:\Users\Amera\Flutter Projects\chap3_part2> flutter pub add flutter_svg
```

Hands-On – Restaurant App

- Open the **pubspec.yaml** file, and add a new dependency, **flutter_svg**
- In the same yaml file, add the location path for your icons.
- Click the “Pub get” on the top to update your project dependencies.

```
29 # versions available, run 'flutter pub outd
30 dependencies:
31   flutter:
32     sdk: flutter
33
34   # The following adds the Cupertino Icons
35   # Use with the CupertinoIcons class for i
36   cupertino_icons: ^1.0.6
37   flutter_svg: ^2.0.10+1
38
39 dev_dependencies:
40   flutter_test:
41     sdk: flutter
42
43   # The "flutter_lints" package below conta
```

```
55 flutter:
56
57   # The following line ensures that
58   # included with your application, s
59   # the material Icons class.
60   uses-material-design: true
61
62   # To add assets to your application
63   assets:
64     - assets/icons/
65
66   # An image asset can refer to one o
67   # https://flutter.dev/assets-and-in
```



Click

STEP 6 : Create _categorySection()

👉 In homepage.dart, inside the HomePage class, type:

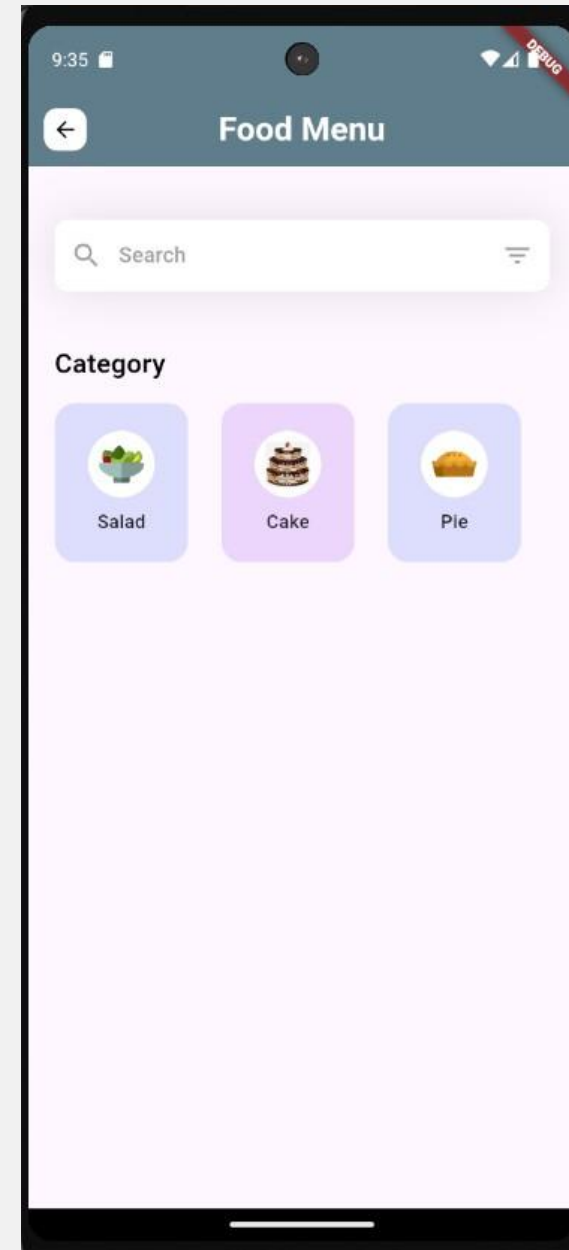
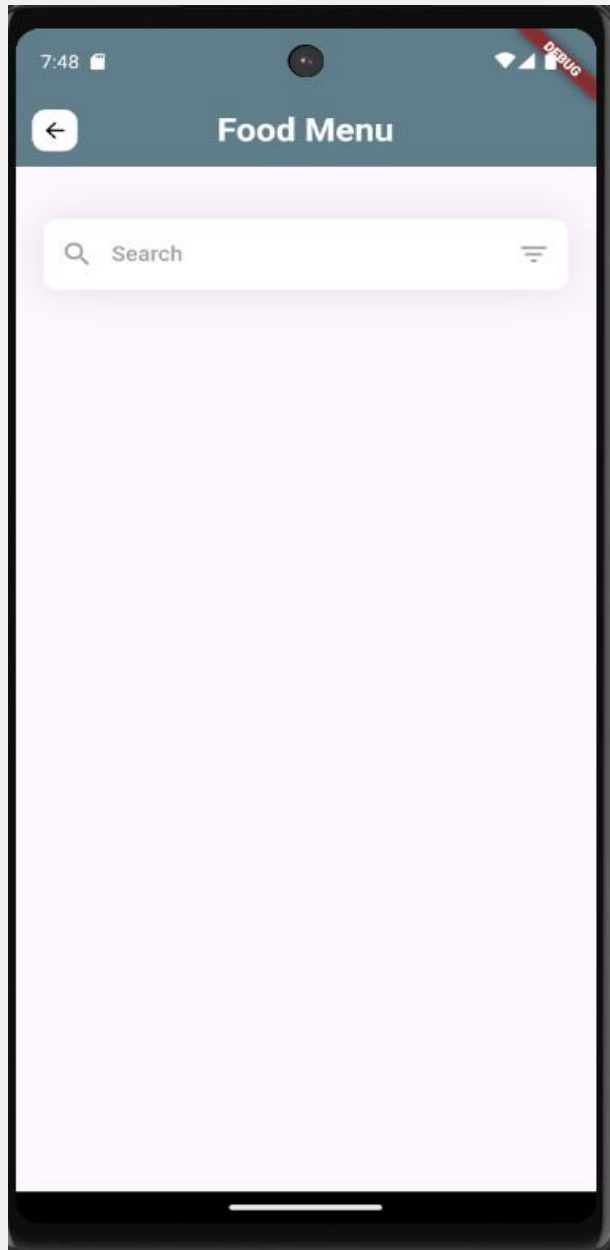
- Copy from categoryList.txt to below SizedBox



```
final List<CategoryModel> categories = CategoryModel.getCategories();
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: _appBarFunction(),
    body: SingleChildScrollView(
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          _searchField(),
          const SizedBox(height: 40),
          _categorySection(),
        ],
      ), // Column
    ), // SingleChildScrollView
  ); // Scaffold
}
```

```
Column _categorySection() {
  return const Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Padding(
        padding: EdgeInsets.only(left: 20),
        child: Text(
          "Category",
          style: TextStyle(
            color: Colors.black,
            fontSize: 20,
            fontWeight: FontWeight.w600,
          ), // TextStyle
        ), // Text
      ), // Padding
      SizedBox(height: 15),
    ],
  ); // Column
}
```



STEP 7 : Create _dietSection()

👉 In homepage.dart, inside the HomePage class, type:

- Copy from dietSection.txt to below SizedBox

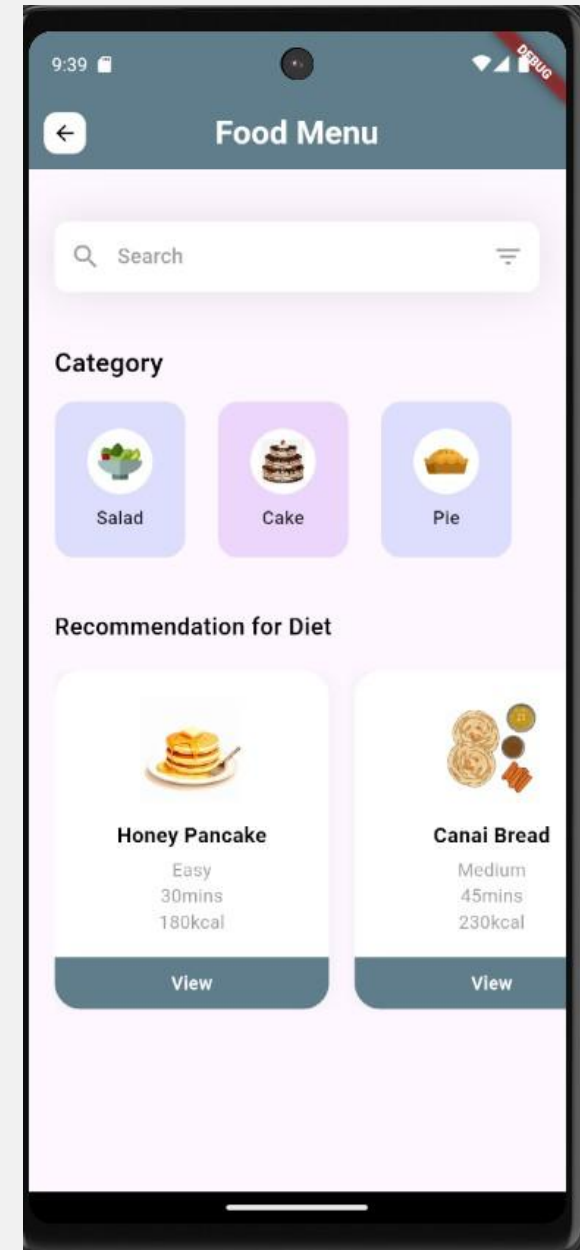
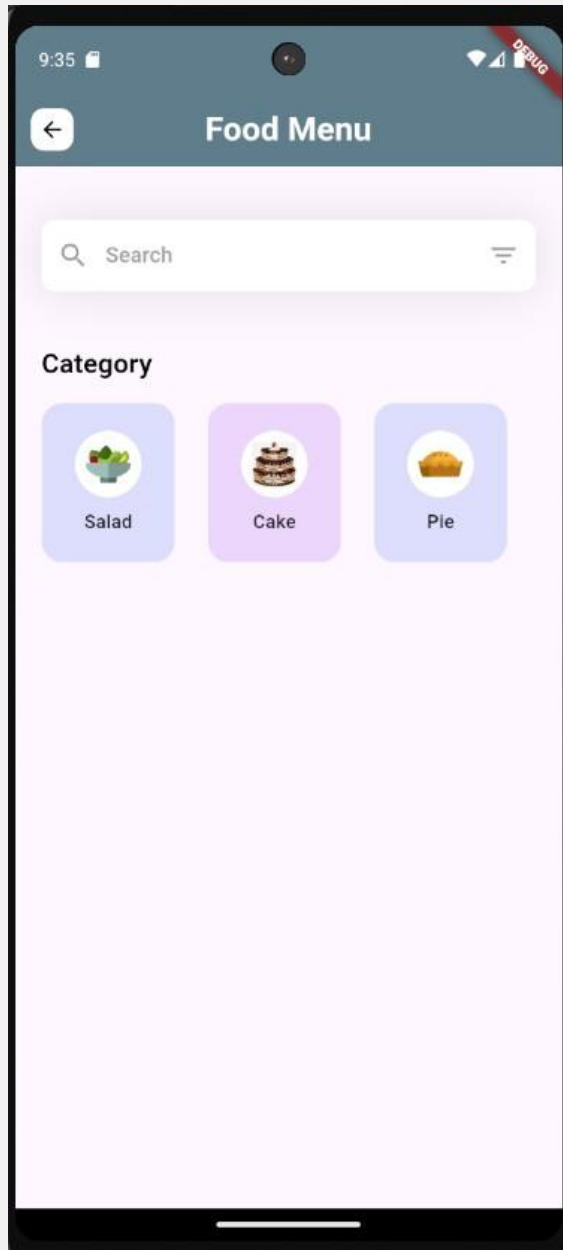


```
class HomePage extends StatelessWidget {  
  HomePage({super.key});  
  
  final List<CategoryModel> categories = CategoryModel.getCategories();  
  final List<DietModel> diets = DietModel.getDiets();  

```

```
    _searchField(),  
    const SizedBox(height: 40),  
    _categorySection(),  
    const SizedBox(height: 40),  
    _dietSection(),  
  ],  
), // Column  
, // SingleChildScrollView  
); // Scaffold  
}
```

```
Column _dietSection() {  
  return const Column(  
    crossAxisAlignment: CrossAxisAlignment.start,  
    children: [  
      Padding(  
        padding: EdgeInsets.only(left: 20),  
        child: Text(  
          "Recommendation for Diet",  
          style: TextStyle(  
            color: Colors.black,  
            fontSize: 18,  
            fontWeight: FontWeight.w600,  
          ), // TextStyle  
        ), // Text  
      ), // Padding  
      SizedBox(height: 20),  
    ],  
  ); // Column  
}
```



See you next class